

1. Automated Dataset Staging

AlphaDent dataset
(Kaggle mount)

Zenodo mirror
(fallback)

Recursive copy to writable dir
`shutil.copytree(...) → labels, images, YAML`

2. YOLO26 Training (Segment26 head)

Dynamic GPU enumeration
 $\mathcal{D} = \{0, \dots, N_{\text{GPU}} - 1\}$

DDP multi-GPU training ($\approx 1.85 \times$ speedup)
YOLO26m@640 / YOLO26l@960 / YOLO26x@960

3. Memory-Safe Validation

Single device $\mathcal{D}_0$; disable raw dumps;
`gc.collect() + empty_cache()`

Self-healing fallback:
CUDA OOM $\Rightarrow$ re-run `val()` on CPU

4. Prediction & Evaluation

9-class pathology instance masks
Box / Mask mAP@50 metrics

5. Explainability (XAI)

YOLOExplainer: hooks on C3k2 neck layer (idx 22)
 $\rightarrow$ activations A^k ; Segment26 class-score
backprop $\rightarrow$ gradients $\partial y^c / \partial A^k$

7 CAM methods: Eigen · Grad · Grad++
· XGrad · HiRes · Layer · EigenGrad

Clinician visual audit of lesion attributions